

1. PHÂN HỆ ĐỊNH DANH & PHÂN QUYỀN (Identity & RBAC)

ROLES		
PK	id	BIGINT?
	code	NVARCHAR(50) UQ
	name	NVARCHAR(100)
	description	NVARCHAR(255)?

PERMISSIONS		
PK	id	BIGINT?
	code	NVARCHAR(100) UQ
	name	NVARCHAR(150)
	module	NVARCHAR(50)

ROLE_PERMISSIONS		
PK	id	BIGINT?
FK	role_id	BIGINT
FK	permission_id	BIGINT

USERS		
PK	id	BIGINT?
FK	role_id	BIGINT
	full_name	NVARCHAR(150)
	email	NVARCHAR(150) UQ
	phone	NVARCHAR(20) UQ?
	password_hash	NVARCHAR(255)
	dob	DATE?
	gender	NVARCHAR(10)?
	avatar_url	NVARCHAR(255)?
	address	NVARCHAR(255)?
	status	NVARCHAR(20)
	last_login_at	DATETIME2?
	updated_at	DATETIME2
	deleted_at	DATETIME2?

MEMBERS		
PK,FK	user_id	BIGINT?
	membership_code	NVARCHAR(30) UQ
	join_date	DATE
	health_notes	NVARCHAR(MAX)?
	fitness_goal	NVARCHAR(255)?
	fitness_level	NVARCHAR(20)
	emergency_contact_name	NVARCHAR(150)?
	emergency_contact_phone	NVARCHAR(20)?

COACHES		
PK,FK	user_id	BIGINT?
	specialization	NVARCHAR(150)?
	bio	NVARCHAR(MAX)?
	certification	NVARCHAR(255)?
	hire_date	DATE?
	employment_status	NVARCHAR(20)

RECEPTIONISTS		
PK,FK	user_id	BIGINT?
	hire_date	DATE?
	shift	NVARCHAR(50)?
	employment_status	NVARCHAR(20)

CENTER MANAGERS		
PK,FK	user_id	BIGINT?
	hire_date	DATE?

3. PHÂN HỆ ĐĂNG KÝ, GIAO DỊCH & HÓA ĐƠN (Subscriptions & Invoicing)

MEMBERSHIP_PACKAGES		
PK	id	BIGINT?
	name	NVARCHAR(150)
	description	NVARCHAR(MAX)?
	price	DECIMAL(12)
	duration_days	INT
	class_credit_limit	INT?
	status	NVARCHAR(20)
	updated_at	DATETIME2

MEMBERSHIP_SUBSCRIPTIONS		
PK	id	BIGINT?
FK	member_id	BIGINT
FK	package_id	BIGINT
FK	previous_subscription_id	BIGINT?
	start_date	DATE
	end_date	DATE
	status	NVARCHAR(20)
	qr_code	NVARCHAR(255) UQ?
	updated_at	DATETIME2

PAYMENTS		
PK	id	BIGINT?
FK	member_id	BIGINT
FK	subscription_id	BIGINT?
FK	class_enrollment_id	BIGINT?
	amount	DECIMAL(12)
	method	NVARCHAR(20)
	status	NVARCHAR(20)
	paid_at	DATETIME2?
FK	received_by	BIGINT?
	note	NVARCHAR(255)?

REPORT_SNAPSHOTS		
PK	id	BIGINT?
	report_type	NVARCHAR(30)
	period_start	DATE
	period_end	DATE
	data	NVARCHAR(MAX)
FK	generated_by	BIGINT?
	generated_at	DATETIME2

INVOICES		
PK	id	BIGINT?
FK	payment_id	BIGINT UQ
	invoice_number	NVARCHAR(50) UQ
	issued_at	DATETIME2
	subtotal_amount	DECIMAL(12)
	tax_amount	DECIMAL(12)
	total_amount	COMPUTED
	pdf_url	NVARCHAR(255)?

INVOICE_ITEMS		
PK	id	BIGINT?
FK	invoice_id	BIGINT
	description	NVARCHAR(255)
	quantity	INT
	unit_price	DECIMAL(12)
	amount	COMPUTED

CENTER_CHECKINS		
PK	id	BIGINT?
FK	member_id	BIGINT
	check_in_time	DATETIME2
	check_out_time	DATETIME2?
	method	NVARCHAR(10)
FK	recorded_by	BIGINT?
	gate	NVARCHAR(50)?

2. PHÂN HỆ CƠ SỞ VẬT CHẤT & GÓI TẬP (Facilities & Packages)

DISCIPLINES		
PK	id	BIGINT?
	name	NVARCHAR(100) UQ
	description	NVARCHAR(MAX)?

ROOMS		
PK	id	BIGINT?
	name	NVARCHAR(100)
	location	NVARCHAR(150)?
	capacity	INT
	status	NVARCHAR(20)

4. PHÂN HỆ LỚP HỌC, LỊCH TẬP & ĐIỂM DANH (Classes & Attendance)

CLASSES		
PK	id	BIGINT?
	name	NVARCHAR(150)
FK	discipline_id	BIGINT
FK	coach_id	BIGINT
FK	room_id	BIGINT
	capacity	INT
	level	NVARCHAR(20)
	description	NVARCHAR(MAX)?
	status	NVARCHAR(20)
	updated_at	DATETIME2

CLASS_ENROLLMENTS		
PK	id	BIGINT?
FK	class_id	BIGINT
FK	member_id	BIGINT
	status	NVARCHAR(20)
	enrolled_at	DATETIME2
	cancelled_at	DATETIME2?
FK	cancelled_by	BIGINT?
	cancel_reason	NVARCHAR(255)?

CLASS_WAITLISTS		
PK	id	BIGINT?
FK	class_id	BIGINT
FK	member_id	BIGINT
	requested_at	DATETIME2
	status	NVARCHAR(20)

CLASS_SESSIONS		
PK	id	BIGINT?
FK	class_id	BIGINT
	session_date	DATE
	start_time	TIME
	end_time	TIME
	status	NVARCHAR(20)
	cancel_reason	NVARCHAR(255)?
	updated_at	DATETIME2

SESSION_ATTENDANCE		
PK	id	BIGINT?
FK	session_id	BIGINT
FK	member_id	BIGINT
	status	NVARCHAR(20)
	checked_in_at	DATETIME2?
FK	recorded_by	BIGINT
	notes	NVARCHAR(255)?

SESSION_EVALUATIONS		
PK	id	BIGINT?
FK	session_id	BIGINT?
FK	member_id	BIGINT
FK	coach_id	BIGINT
	performance_notes	NVARCHAR(MAX)?
	progress_score	DECIMAL(4)?
	feedback	NVARCHAR(MAX)?
	evaluated_at	DATETIME2

5. PHÂN HỆ HUẤN LUYỆN & THEO DÕI TIẾN ĐỘ (Coaching & Progress)

TRAINING_PLANS		
PK	id	BIGINT?
FK	member_id	BIGINT
FK	coach_id	BIGINT
FK	class_id	BIGINT?
	goal	NVARCHAR(255)?
	start_date	DATE
	end_date	DATE
	status	NVARCHAR(20)
	is_ai_generated	BIT
FK	ai_recommendation_id	BIGINT?
	updated_at	DATETIME2

TRAINING_PLAN_ITEMS		
PK	id	BIGINT?
FK	training_plan_id	BIGINT
	day_of_week	TINYINT?
	exercise_name	NVARCHAR(150)
	description	NVARCHAR(MAX)?
	sets	INT?
	reps	INT?
	duration_minutes	INT?
	order_index	INT

MEMBER_PROGRESS_LOGS		
PK	id	BIGINT?
FK	member_id	BIGINT
	metric_name	NVARCHAR(100)
	metric_value	DECIMAL(10)
	unit	NVARCHAR(20)?
FK	recorded_by	BIGINT
	recorded_at	DATETIME2

6. PHÂN HỆ TRỢ LÝ AI & CHATBOT (AI Engine & Chat)

AI_RECOMMENDATION_LOGS		
PK	id	BIGINT?
FK	coach_id	BIGINT?
FK	member_id	BIGINT
	input_context	NVARCHAR(MAX)
	recommended_content	NVARCHAR(MAX)
	model_name	NVARCHAR(100)?
	is_applied	BIT

AI_CHAT_SESSIONS		
PK	id	BIGINT?
FK	member_id	BIGINT
	topic	NVARCHAR(150)?
	started_at	DATETIME2
	ended_at	DATETIME2?

AI_CHAT_MESSAGES		
PK	id	BIGINT?
FK	session_id	BIGINT
	sender	NVARCHAR(10)
	message	NVARCHAR(MAX)

7. PHÂN HỆ THÔNG BÁO, HỖ TRỢ & NHẬT KÝ (Support & Audit)

SUPPORT_REQUESTS		
PK	id	BIGINT?
FK	member_id	BIGINT
FK	assigned_to	BIGINT?
	subject	NVARCHAR(150)
	description	NVARCHAR(MAX)?
	status	NVARCHAR(20)
	resolved_at	DATETIME2?

SUPPORT_REQUEST_MESSAGES		
PK	id	BIGINT?
FK	support_request_id	BIGINT
FK	sender_user_id	BIGINT
	message	NVARCHAR(MAX)

NOTIFICATIONS		
PK	id	BIGINT?
FK	user_id	BIGINT
	type	NVARCHAR(30)
	title	NVARCHAR(150)
	message	NVARCHAR(MAX)
	is_read	BIT
	related_entity_type	NVARCHAR(50)?
	related_entity_type	IS?
	related_entity_id	BIGINT?

AUDIT_LOGS		
PK	id	BIGINT?
FK	user_id	BIGINT?
	action	NVARCHAR(100)
	entity_type	NVARCHAR(100)
	entity_id	BIGINT?
	old_value	NVARCHAR(MAX)?
	new_value	NVARCHAR(MAX)?
	ip_address	NVARCHAR(45)?

ROLES		
PK	id	BIGINT?
	code	NVARCHAR(50) UQ
	name	NVARCHAR(100)
	description	NVARCHAR(255)?

PERMISSIONS		
PK	id	BIGINT?
	code	NVARCHAR(100) UQ
	name	NVARCHAR(150)
	module	NVARCHAR(50)

ROLE_PERMISSIONS		
PK	id	BIGINT?
FK	role_id	BIGINT
FK	permission_id	BIGINT

USERS		
PK	id	BIGINT?
FK	role_id	BIGINT
	full_name	NVARCHAR(150)
	email	NVARCHAR(150) UQ
	phone	NVARCHAR(20) UQ?
	password_hash	NVARCHAR(255)
	dob	DATE?
	gender	NVARCHAR(10)?
	avatar_url	NVARCHAR(255)?
	address	NVARCHAR(255)?
	status	NVARCHAR(20)
	last_login_at	DATETIME2?
	updated_at	DATETIME2
	deleted_at	DATETIME2?

CENTER MANAGERS		
PK,FK	user_id	BIGINT?
	hire_date	DATE?

MEMBERS		
PK,FK	user_id	BIGINT?
	membership_code	NVARCHAR(30) UQ
	join_date	DATE
	health_notes	NVARCHAR(MAX)?
	fitness_goal	NVARCHAR(255)?
	fitness_level	NVARCHAR(20)
	emergency_contact_name	NVARCHAR(150)?
	emergency_contact_phone	NVARCHAR(20)?

COACHES		
PK,FK	user_id	BIGINT?
	specialization	NVARCHAR(150)?
	bio	NVARCHAR(MAX)?
	certification	NVARCHAR(255)?
	hire_date	DATE?
	employment_status	NVARCHAR(20)

RECEPTIONISTS		
PK,FK	user_id	BIGINT?
	hire_date	DATE?
	shift	NVARCHAR(50)?
	employment_status	NVARCHAR(20)

USERS		
PK	id	BIGINT?
FK	role_id	BIGINT
	full_name	NVARCHAR(150)
	email	NVARCHAR(150) UQ
	phone	NVARCHAR(20) UQ?
	password_hash	NVARCHAR(255)
	dob	DATE?
	gender	NVARCHAR(10)?
	avatar_url	NVARCHAR(255)?
	address	NVARCHAR(255)?
	status	NVARCHAR(20)
	last_login_at	DATETIME2?
	updated_at	DATETIME2
	deleted_at	DATETIME2?

MEMBERS		
PK,FK	user_id	BIGINT?
	membership_code	NVARCHAR(30) UQ
	join_date	DATE
	health_notes	NVARCHAR(MAX)?
	fitness_goal	NVARCHAR(255)?
	fitness_level	NVARCHAR(20)
	emergency_contact_name	NVARCHAR(150)?
	emergency_contact_phone	NVARCHAR(20)?

MEMBERSHIP_PACKAGES		
PK	id	BIGINT?
	name	NVARCHAR(150)
	description	NVARCHAR(MAX)?
	price	DECIMAL(12
	duration_days	INT
	class_credit_limit	INT?
	status	NVARCHAR(20)
	updated_at	DATETIME2

MEMBERSHIP_SUBSCRIPTIONS		
PK	id	BIGINT?
FK	member_id	BIGINT
FK	package_id	BIGINT
FK	previous_subscription_id	BIGINT?
	start_date	DATE
	end_date	DATE
	status	NVARCHAR(20)
	qr_code	NVARCHAR(255) UQ?
	updated_at	DATETIME2

REPORT_SNAPSHOTS		
PK	id	BIGINT?
	report_type	NVARCHAR(30)
	period_start	DATE
	period_end	DATE
	data	NVARCHAR(MAX)
FK	generated_by	BIGINT?
	generated_at	DATETIME2

PAYMENTS		
PK	id	BIGINT?
FK	member_id	BIGINT
FK	subscription_id	BIGINT?
FK	class_enrollment_id	BIGINT?
	amount	DECIMAL(12
	method	NVARCHAR(20)
	status	NVARCHAR(20)
	paid_at	DATETIME2?
FK	received_by	BIGINT?
	note	NVARCHAR(255)?

INVOICES		
PK	id	BIGINT?
FK	payment_id	BIGINT UQ
	invoice_number	NVARCHAR(50) UQ
	issued_at	DATETIME2
	subtotal_amount	DECIMAL(12
	tax_amount	DECIMAL(12
	total_amount	COMPUTED
	pdf_url	NVARCHAR(255)?

INVOICE_ITEMS		
PK	id	BIGINT?
FK	invoice_id	BIGINT
	description	NVARCHAR(255)
	quantity	INT
	unit_price	DECIMAL(12
	amount	COMPUTED

DISCIPLINES		
PK	id	BIGINT?
	name	NVARCHAR(100) UQ
	description	NVARCHAR(MAX)?

ROOMS		
PK	id	BIGINT?
	name	NVARCHAR(100)
	location	NVARCHAR(150)?
	capacity	INT
	status	NVARCHAR(20)

COACHES		
PK,FK	user_id	BIGINT?
	specialization	NVARCHAR(150)?
	bio	NVARCHAR(MAX)?
	certification	NVARCHAR(255)?
	hire_date	DATE?
	employment_status	NVARCHAR(20)

CLASSES		
PK	id	BIGINT?
	name	NVARCHAR(150)
FK	discipline_id	BIGINT
FK	coach_id	BIGINT
FK	room_id	BIGINT
	capacity	INT
	level	NVARCHAR(20)
	description	NVARCHAR(MAX)?
	status	NVARCHAR(20)
	updated_at	DATETIME2

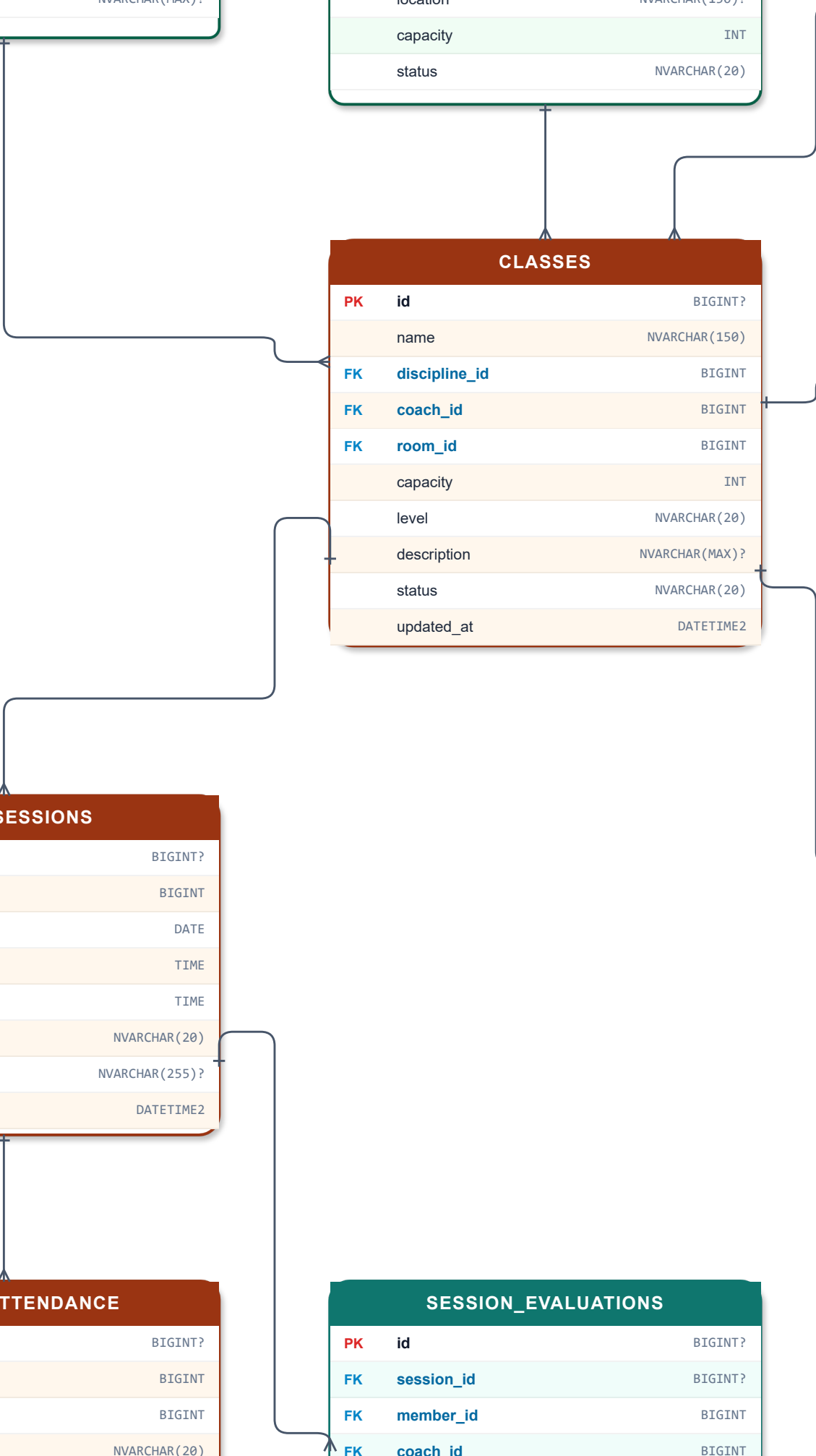
CLASS_ENROLLMENTS		
PK	id	BIGINT?
FK	class_id	BIGINT
FK	member_id	BIGINT
	status	NVARCHAR(20)
	enrolled_at	DATETIME2
	cancelled_at	DATETIME2?
FK	cancelled_by	BIGINT?
	cancel_reason	NVARCHAR(255)?

CLASS_SESSIONS		
PK	id	BIGINT?
FK	class_id	BIGINT
	session_date	DATE
	start_time	TIME
	end_time	TIME
	status	NVARCHAR(20)
	cancel_reason	NVARCHAR(255)?
	updated_at	DATETIME2

CLASS_WAITLISTS		
PK	id	BIGINT?
FK	class_id	BIGINT
FK	member_id	BIGINT
	requested_at	DATETIME2
	status	NVARCHAR(20)

SESSION_ATTENDANCE		
PK	id	BIGINT?
FK	session_id	BIGINT
FK	member_id	BIGINT
	status	NVARCHAR(20)
	checked_in_at	DATETIME2?
FK	recorded_by	BIGINT
	notes	NVARCHAR(255)?

SESSION_EVALUATIONS		
PK	id	BIGINT?
FK	session_id	BIGINT?
FK	member_id	BIGINT
FK	coach_id	BIGINT
	performance_notes	NVARCHAR(MAX)?
	progress_score	DECIMAL(4?
	feedback	NVARCHAR(MAX)?
	evaluated_at	DATETIME2



COACHES		
PK,FK	user_id	BIGINT?
	specialization	NVARCHAR(150)?
	bio	NVARCHAR(MAX)?
	certification	NVARCHAR(255)?
	hire_date	DATE?
	employment_status	NVARCHAR(20)

MEMBERS		
PK,FK	user_id	BIGINT?
	membership_code	NVARCHAR(30) UQ
	join_date	DATE
	health_notes	NVARCHAR(MAX)?
	fitness_goal	NVARCHAR(255)?
	fitness_level	NVARCHAR(20)
	emergency_contact_name	NVARCHAR(150)?
	emergency_contact_phone	NVARCHAR(20)?

AI_CHAT_SESSIONS		
PK	id	BIGINT?
FK	member_id	BIGINT
	topic	NVARCHAR(150)?
	started_at	DATETIME2
	ended_at	DATETIME2?

AI_CHAT_MESSAGES		
PK	id	BIGINT?
FK	session_id	BIGINT
	sender	NVARCHAR(10)
	message	NVARCHAR(MAX)

SESSION_EVALUATIONS		
PK	id	BIGINT?
FK	session_id	BIGINT?
FK	member_id	BIGINT
FK	coach_id	BIGINT
	performance_notes	NVARCHAR(MAX)?
	progress_score	DECIMAL(4?
	feedback	NVARCHAR(MAX)?
	evaluated_at	DATETIME2

TRAINING_PLANS		
PK	id	BIGINT?
FK	member_id	BIGINT
FK	coach_id	BIGINT
FK	class_id	BIGINT?
	goal	NVARCHAR(255)?
	start_date	DATE
	end_date	DATE
	status	NVARCHAR(20)
	is_ai_generated	BIT
FK	ai_recommendation_id	BIGINT?
	updated_at	DATETIME2

AI_RECOMMENDATION_LOGS		
PK	id	BIGINT?
FK	coach_id	BIGINT?
FK	member_id	BIGINT
	input_context	NVARCHAR(MAX)
	recommended_content	NVARCHAR(MAX)
	model_name	NVARCHAR(100)?
	is_applied	BIT

TRAINING_PLAN_ITEMS		
PK	id	BIGINT?
FK	training_plan_id	BIGINT
	day_of_week	TINYINT?
	exercise_name	NVARCHAR(150)
	description	NVARCHAR(MAX)?
	sets	INT?
	reps	INT?
	duration_minutes	INT?
	order_index	INT

MEMBER_PROGRESS_LOGS		
PK	id	BIGINT?
FK	member_id	BIGINT
	metric_name	NVARCHAR(100)
	metric_value	DECIMAL(10
	unit	NVARCHAR(20)?
FK	recorded_by	BIGINT
	recorded_at	DATETIME2

USERS		
PK	id	BIGINT?
FK	role_id	BIGINT
	full_name	NVARCHAR(150)
	email	NVARCHAR(150) UQ
	phone	NVARCHAR(20) UQ?
	password_hash	NVARCHAR(255)
	dob	DATE?
	gender	NVARCHAR(10)?
	avatar_url	NVARCHAR(255)?
	address	NVARCHAR(255)?
	status	NVARCHAR(20)
	last_login_at	DATETIME2?
	updated_at	DATETIME2
	deleted_at	DATETIME2?

MEMBERS		
PK,FK	user_id	BIGINT?
	membership_code	NVARCHAR(30) UQ
	join_date	DATE
	health_notes	NVARCHAR(MAX)?
	fitness_goal	NVARCHAR(255)?
	fitness_level	NVARCHAR(20)
	emergency_contact_name	NVARCHAR(150)?
	emergency_contact_phone	NVARCHAR(20)?

SUPPORT_REQUESTS		
PK	id	BIGINT?
FK	member_id	BIGINT
FK	assigned_to	BIGINT?
	subject	NVARCHAR(150)
	description	NVARCHAR(MAX)?
	status	NVARCHAR(20)
	resolved_at	DATETIME2?

SUPPORT_REQUEST_MESSAGES		
PK	id	BIGINT?
FK	support_request_id	BIGINT
FK	sender_user_id	BIGINT
	message	NVARCHAR(MAX)

NOTIFICATIONS		
PK	id	BIGINT?
FK	user_id	BIGINT
	type	NVARCHAR(30)
	title	NVARCHAR(150)
	message	NVARCHAR(MAX)
	is_read	BIT
	related_entity_type	NVARCHAR(50)?
	related_entity_type	IS?
	related_entity_id	BIGINT?

AUDIT_LOGS		
PK	id	BIGINT?
FK	user_id	BIGINT?
	action	NVARCHAR(100)
	entity_type	NVARCHAR(100)
	entity_id	BIGINT?
	old_value	NVARCHAR(MAX)?
	new_value	NVARCHAR(MAX)?
	ip_address	NVARCHAR(45)?